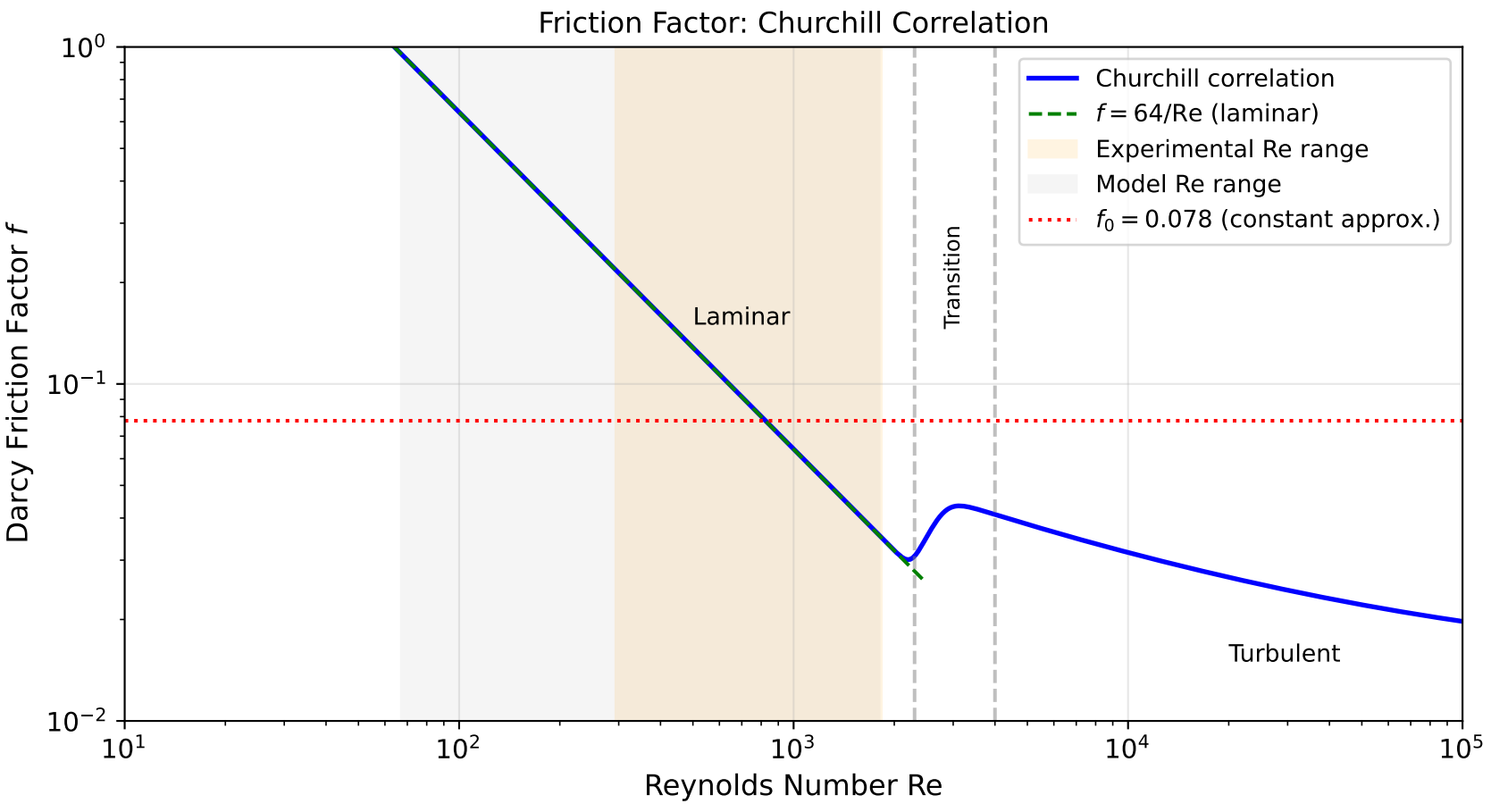
Friction Factor: Churchill Correlation
Darcy Friction Factor f
Reynolds Number Re
10^0
10^{-1}
10^{-2}
10^1
10^2
10^3
10^4
10^5
Churchill correlation
$f = 64/\mathrm{Re}$ (laminar)
Experimental Re range
Model Re range
$f_0 = 0.078$ (constant approx.)
Laminar
Transition
Turbulent